Based on the output obtained, justify if we can reject the null hypothesis? Also, what can you conclude about the relationship between x and y? [4 Marks]

(c) Using R, on the same dataset, now we fit the linear model:

$$y_i = 1.5 + 2x_i + \epsilon_i \quad ; \quad 1 \leq i \leq n$$

The output obtained is as follows:

```
Call:
lm(formula = y ~ x)

Residuals:
    Min      1Q  Median      3Q     Max
-3.9053 -0.7122  0.0545  0.7265  2.3814

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  1.58118    0.11591   13.64   <2e-16 ***
x            1.97353    0.01893  104.26   <2e-16 ***
---
Signif. codes:
0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1.128 on 98 degrees of freedom
Multiple R-squared:  0.9911,    Adjusted R-squared:  0.991
F-statistic: 1.087e+04 on 1 and 98 DF,  p-value: < 2.2e-16
```

Based on the output obtained, justify if we can reject the null hypothesis? Also, what can you conclude about the relationship between x and y? [4 Marks]

d) In which among the above two models, obtained in part (b) and (c), the independent variables collectively explain more variability in the dependent variable? Elaborate on your answer. [4 Marks]

Options :

6406532311818. ✔ I have written answers on the answer sheets

6406532311819. ✖ Not applicable

Speech Technology

Section Id :	64065348601
Section Number :	8
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	3

Number of Questions to be attempted :	3
Section Marks :	25
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653101388
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 131 Question Id : 640653691310 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : SPEECH TECHNOLOGY (PEN AND PAPER EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532311820. ✔ YES

6406532311821. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653101389
Question Shuffling Allowed :	No

Is Section Default? : null

Question Number : 132 Question Id : 640653691311 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 13

Question Label : Multiple Choice Question

1. (1 mark) In a tri−gram language model, the probability of occurrence of a word depends on _____ number of previous words.
   (a) 1
   (b) 2
   (c) 3

2. (3 marks) Find the probability of the sentence: $< s >$ This is a cat $< s >$ with a bi−gram language model trained on sentences below:

   $< s >$ This is a cat $< s >$
   $< s >$ This is a dog $< s >$
   $< s >$ I love my cat $< s >$
   $< s >$ This is my name $< s >$

   Note: $< s >$ is a special token to include start and end words of a sentence in the bi−gram language model. For example, ($< s >$, This) is now considered as one of the bi-grams.

3. (2 marks) Briefly explain the motivation behind the introduction of Connectionist Temporal Classification with an example.

4. (7 points) Consider a weather prediction based HMM problem with three hidden states: *Sunny, Cloudy and Rainy.* There are four different observable states that depend on the hidden states, namely: *Dry, Dryish, Damp, and Soggy.* The transition and observation probabilities are shown in fig. Given initial Cloudy weather condition on Day0. It was observed that the grass was *Dry* followed by *Damp* for the next two days. Compute the best possible weather forecast sequence which best explains these two observations (*hint: Use Viterbi Algorithm*).

Grass

Weather	Dry	Dryish	Damp	Soggy
Sunny	0.60	0.20	0.15	0.05
Cloudy	0.25	0.25	0.25	0.25
Rainy	0.05	0.10	0.35	0.50

Today

Yesterday	Sunny	Cloudy	Rainy
Sunny	0.5	0.375	0.125
Cloudy	0.25	0.125	0.625
Rainy	0.25	0.075	0.675

Options :

6406532311822. ✔ I have written answers on the answer sheets

6406532311823. ✖ Not applicable

Sub-Section Number : 3

Sub-Section Id : 640653101390

Question Shuffling Allowed : No

Is Section Default? : null

Question Number : 133 Question Id : 640653691312 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 12

Question Label : Multiple Choice Question

5. (5 points) Let us assume we want to implement a NAND function using MLP. Below is a tabular representation of NAND function:

X_1	X_2	$X_1 \oplus X_2$
0	0	1
0	1	1
1	0	1
1	1	0

The activation function of a single neuron is denoted as:

$$f(x) = \begin{cases} 1, & \text{for } x < 0 \\ 0, & \text{for } x \geq 0 \end{cases}$$

Model this function using an MLP (*Hint: Use bias for modelling if required*).

6. (3 points) The sigmoid function is given as $f(z) = \frac{1}{1+\exp^{-z}}$. Derive the mathematical expression for the derivative of $f(z)$ in terms of $f(z)$.

7. (2 points) Explain in brief the difference between Markov Chains and Hidden Markov models.

8. (2 points) Suppose you design a multilayer perceptron for classification and it has a single hidden layer with a hard threshold activation function.

$$f(x) = \begin{cases} 0, & \text{for } x < 0 \\ 1, & \text{for } x \geq 0 \end{cases}$$

The output layer uses the softmax activation function with cross-entropy loss. What will go wrong if you try to train this network using gradient descent? Justify your answer in terms of the backpropagation rules. Assume that the weights and biases for initializing the hidden layers have been sampled from a Gaussian with mean 0 and variance 1.

Options :

6406532311824. ✔ I have written answers on the answer sheets

6406532311825. ✖ Not applicable

Advanced Algorithms

Section Id :	64065348602
Section Number :	9
Section type :	Online
Mandatory or Optional :	Mandatory
Number of Questions :	8
Number of Questions to be attempted :	8
Section Marks :	50
Display Number Panel :	Yes
Group All Questions :	No
Enable Mark as Answered Mark for Review and Clear Response :	Yes
Maximum Instruction Time :	0
Sub-Section Number :	1
Sub-Section Id :	640653101391
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Number : 134 Question Id : 640653691313 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DEGREE LEVEL : ADVANCED ALGORITHMS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?